Interactive Imitation Learning

Kianté Brantley

CS 1840 Fall 2026

Announcements

HW4 is out!

HW5 is a mini-project in groups of 2–4.
For HW4, submit your group members and proposed project ideas.

Check the course website for Office Hours:
https://harvard-cs-1840.github.io/2026/

Finishing Up From Last Lecture



1. Offline Imitation Learning: Behavior Cloning

2. Performance difference lemma and its application to proving BC’s bound

Performance Difference Lemma

Given two policies $\pi : S \mapsto \Delta(A),\ \pi' : S \mapsto \Delta(A)$, recall $V^{\pi}(s_0) = \mathbb{E}\left[\sum_{h=0}^{\infty} \gamma^h r(s_h, a_h) \,|\, \pi\right]$

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Performance Difference Lemma (PDL):

$$V^{\pi}(s_0) - V^{\pi'}(s_0) = \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi}_{s_0}}\left[\mathbb{E}_{a\sim\pi(\cdot|s)}Q^{\pi'}(s,a) - V^{\pi'}(s)\right]$$

$$:= \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi}_{s_0}}\left[\mathbb{E}_{a\sim\pi(\cdot|s)}A^{\pi'}(s,a)\right]$$

Analysis of BC

Theorem [BC Performance] With probability at least $1 - \delta$, BC returns a policy $\widehat{\pi}$:

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The quadratic amplification is annoying;
Related to pre-trained LLM hallucination;
Will see how to fix it next lecture

Recap

The Behavior Cloning algorithm:

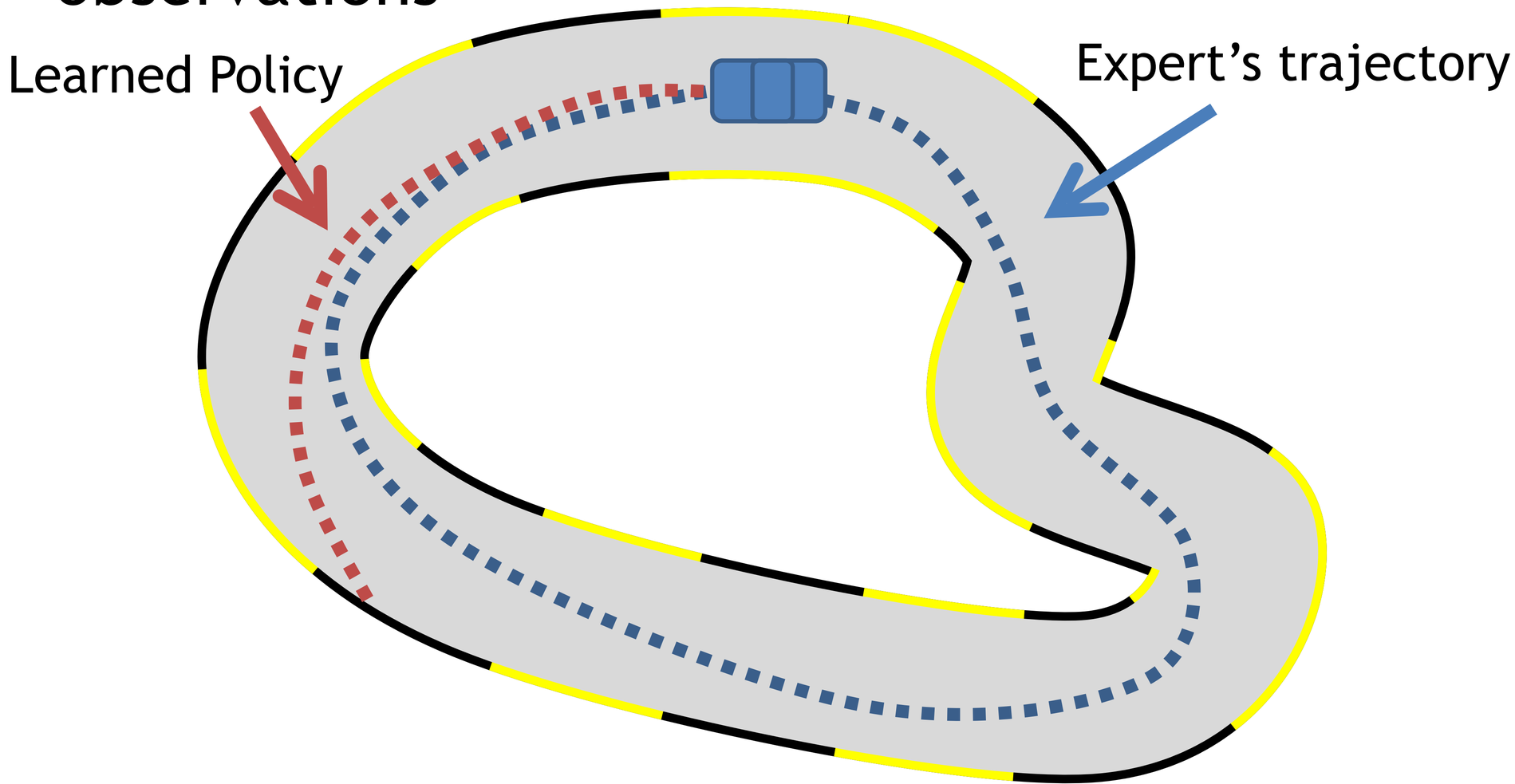
Choose regression (for continuous action) or classification loss $\ell(\pi(s), a)$, and perform SL:

$$\widehat{\pi} = \min_{\pi \in \Pi} \sum_{i=1}^{M} \ell(\pi(s_i^\star), a_i^\star)$$

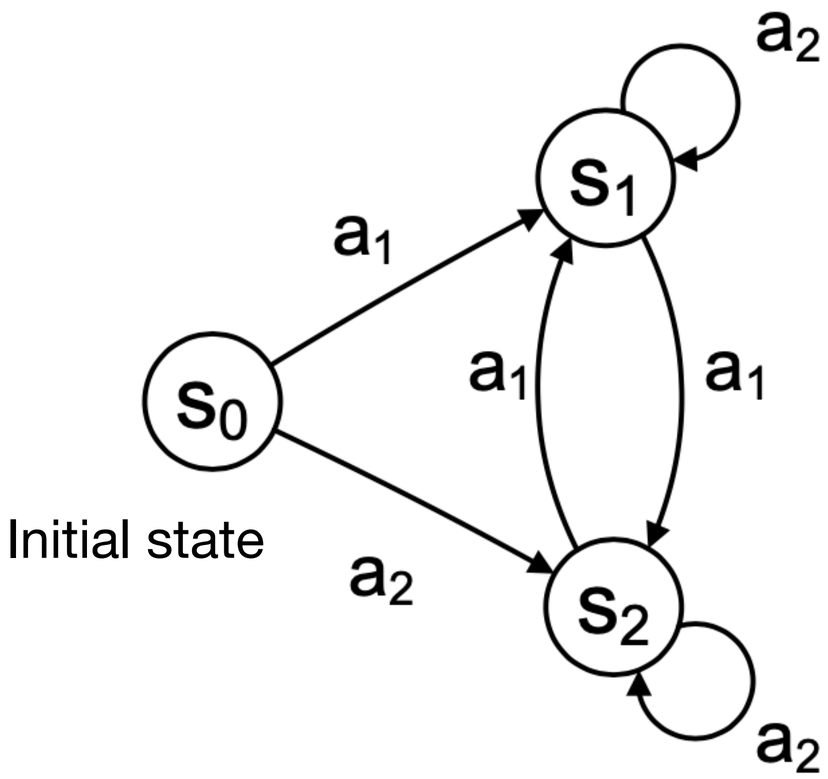
What could go wrong?

[Pomerleau89,Daume09]

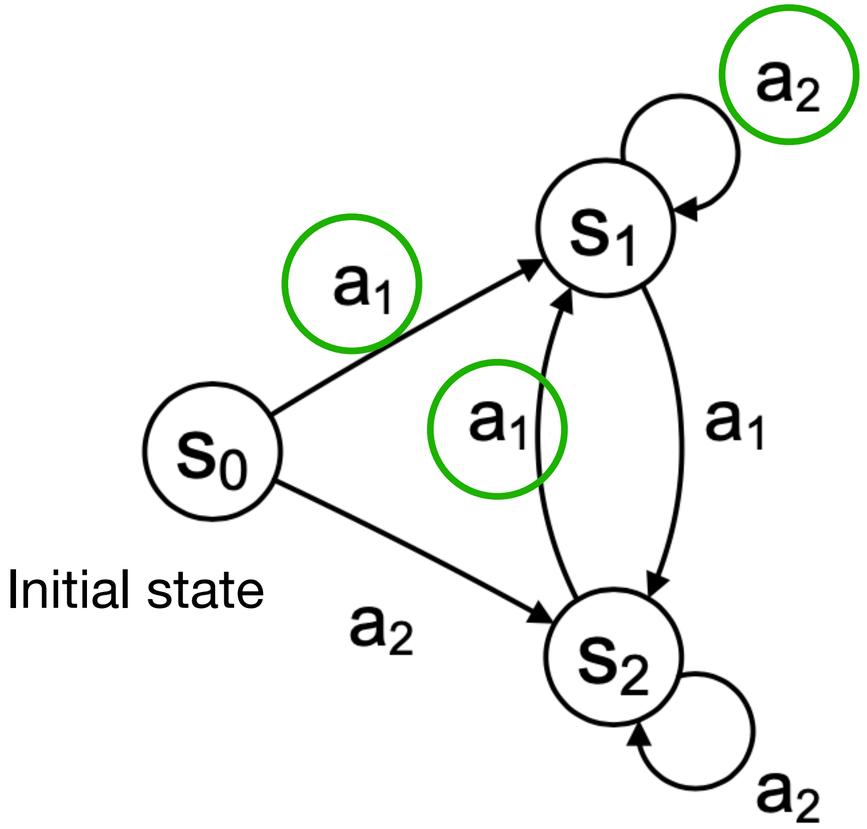
- Predictions affect future inputs/ observations



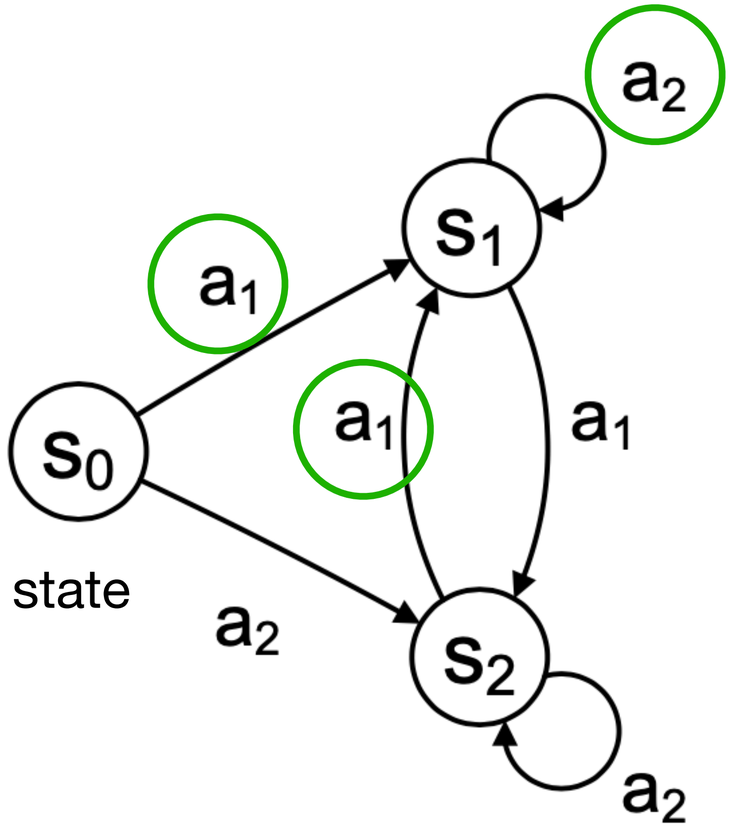
Distribution Shift: Example



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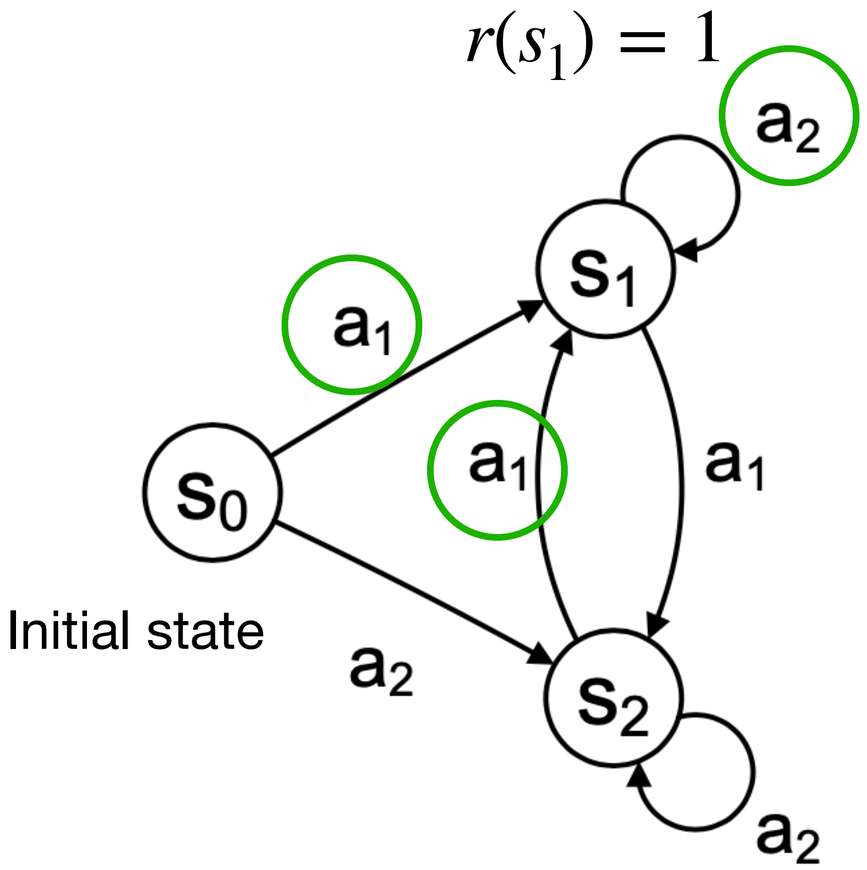


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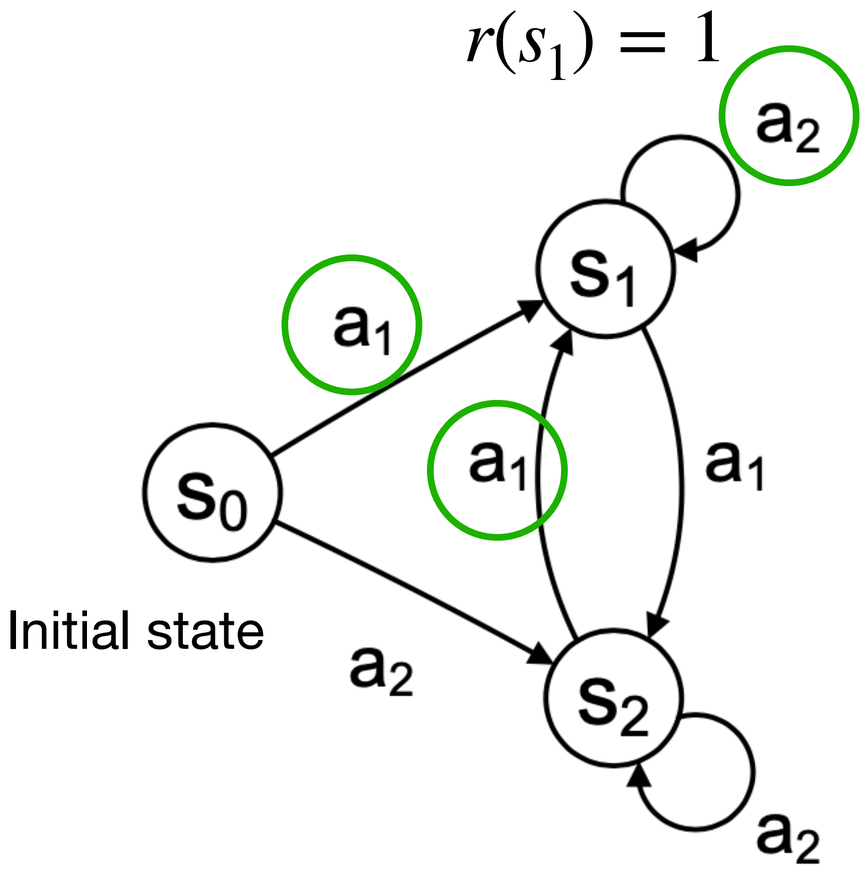
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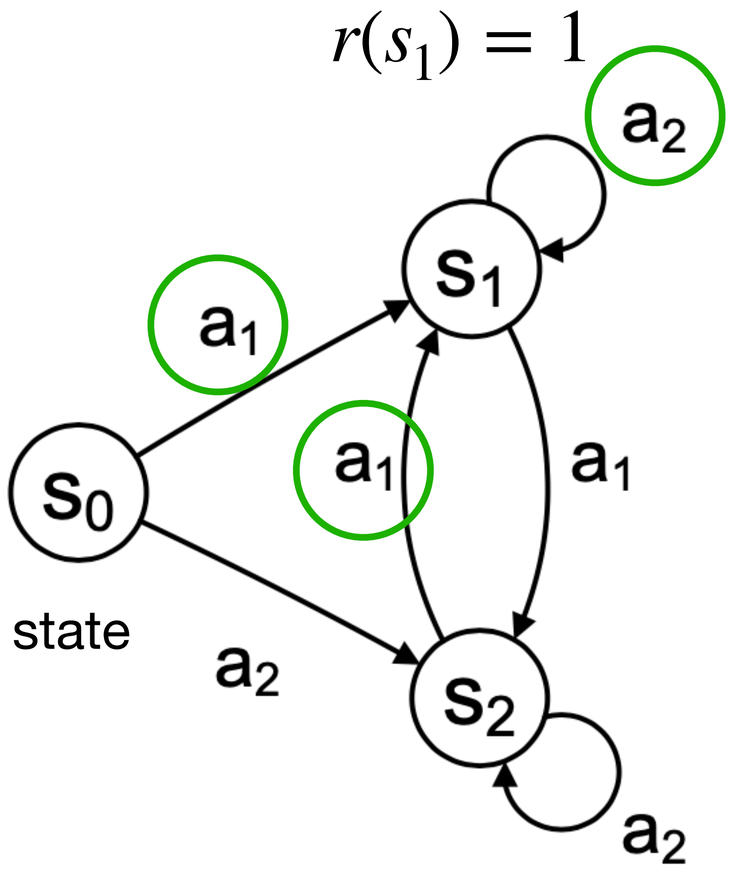
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$$V_{s_0}^{\pi^\star} = \frac{\gamma}{1-\gamma}$$

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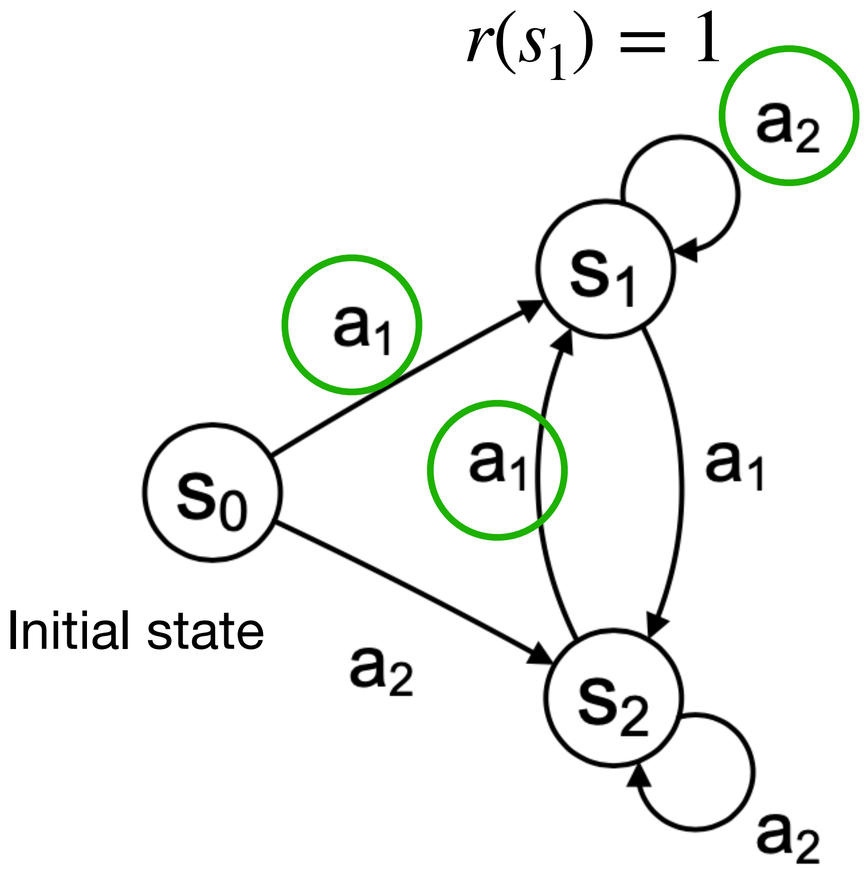
Assume SL returned such policy $\widehat{\pi}$

$$\widehat{\pi}(s_0) = \begin{cases} a_1 & \text{w/ prob } 1 - \epsilon/(1-\gamma) \\ a_2 & \text{w/ prob } \epsilon/(1-\gamma) \end{cases}, \quad \widehat{\pi}(s_1) = a_2, \widehat{\pi}(s_2) = a_2$$

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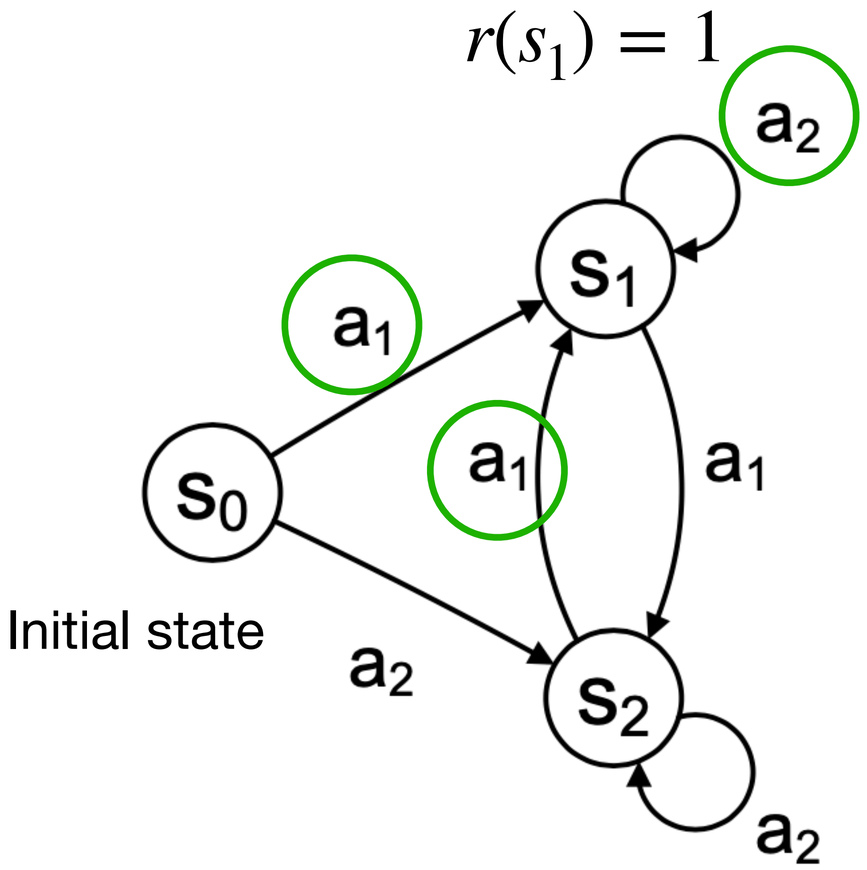
We will have good supervised learning error:

$$\mathbb{E}_{s \sim d_{s_0}^{\pi^\star}} \mathbb{E}_{a \sim \widehat{\pi}(\cdot|s)} \mathbf{1}\left(a \neq \pi^\star(s)\right) = \epsilon$$

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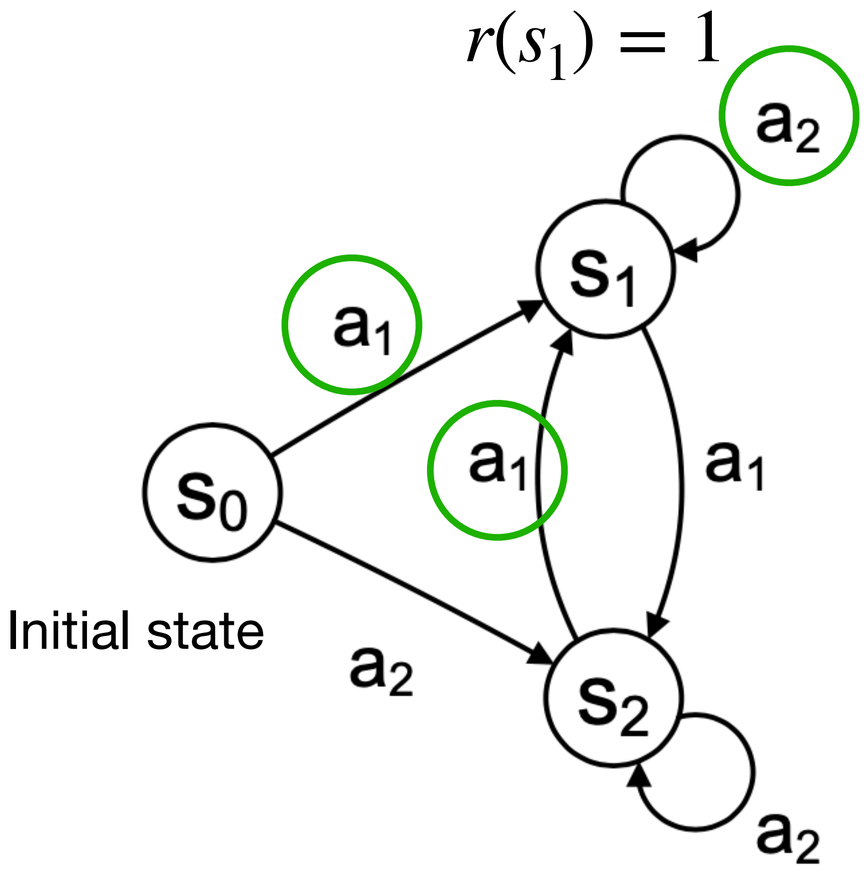
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But we have quadratic error in performance:

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Issue: once we make a mistake at s_0, we end up in s_2 which is not in the training data!

An Autonomous Land Vehicle

In A Neural Network [Pomerleau, NIPS ‘88]



“If the network is not presented with sufficient variability in its training exemplars to cover the conditions it is likely to encounter…[it] will perform poorly”

Question for today:

How to mitigate the distribution shift issue?

Solution:

Interactive Imitation Learning Setting

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Key assumption:

we can query expert $\pi^\star$ at any time and any state during training

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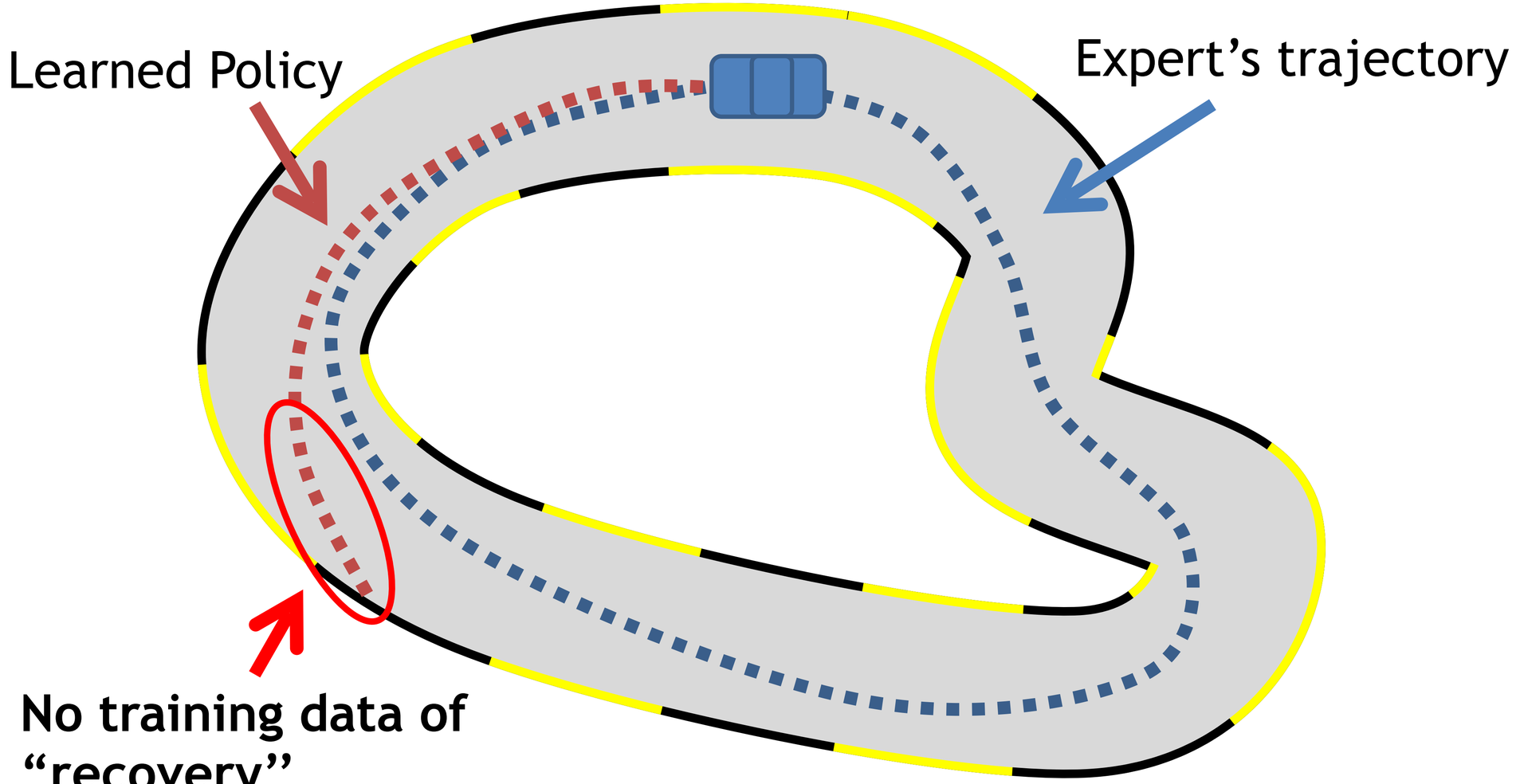
(Recall that previously we only had an offline dataset $\mathscr{D} = (s_i^\star, a_i^\star)_{i=1}^M \sim d_\mu^{\pi^\star}$)

Outline:

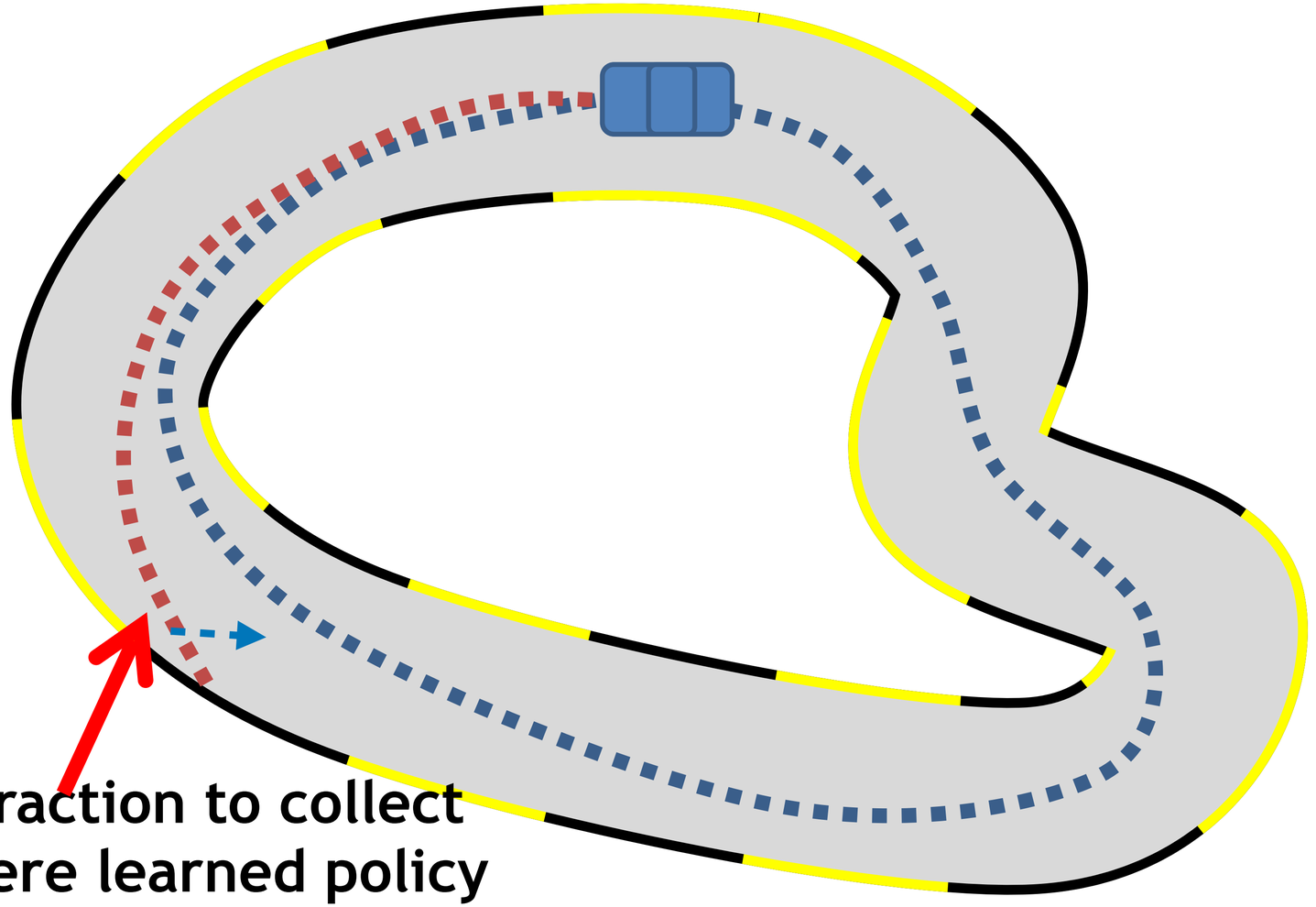
1. The DAgger (Data Aggregation) Algorithm

2. Analysis of DAgger: DAgger as online learning

Recall the Main Problem from Behavior Cloning:

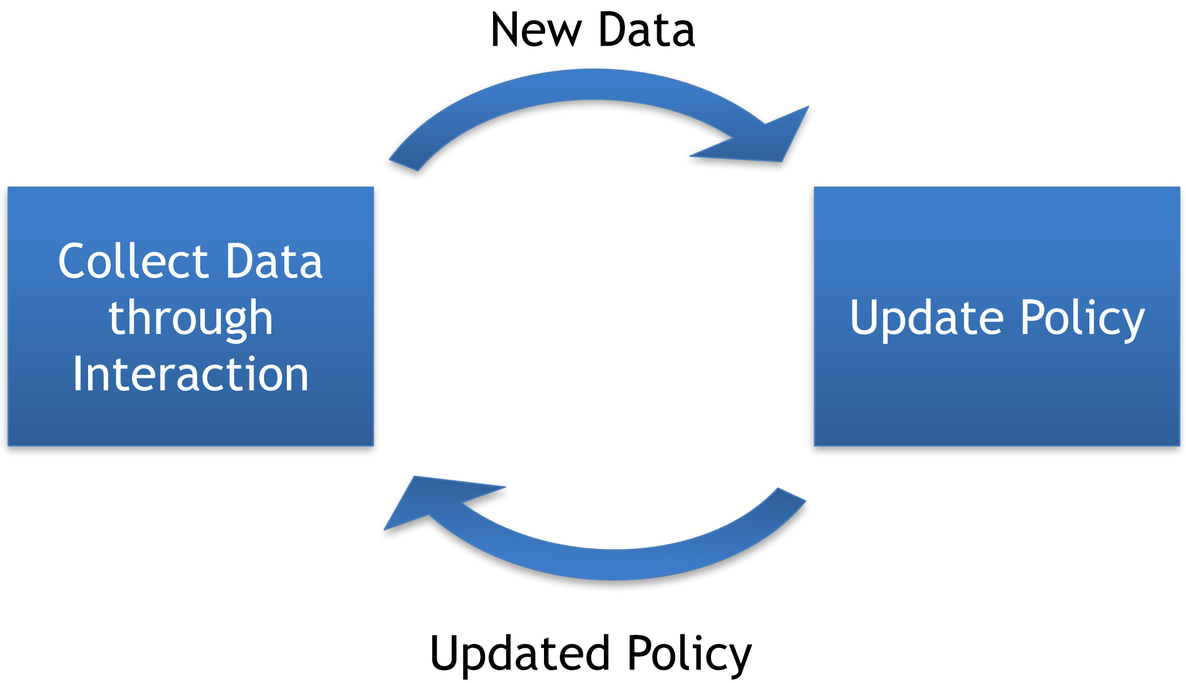


Intuitive solution: Interaction



Use interaction to collect data where learned policy goes

General Idea: Iterative Interactive Approach

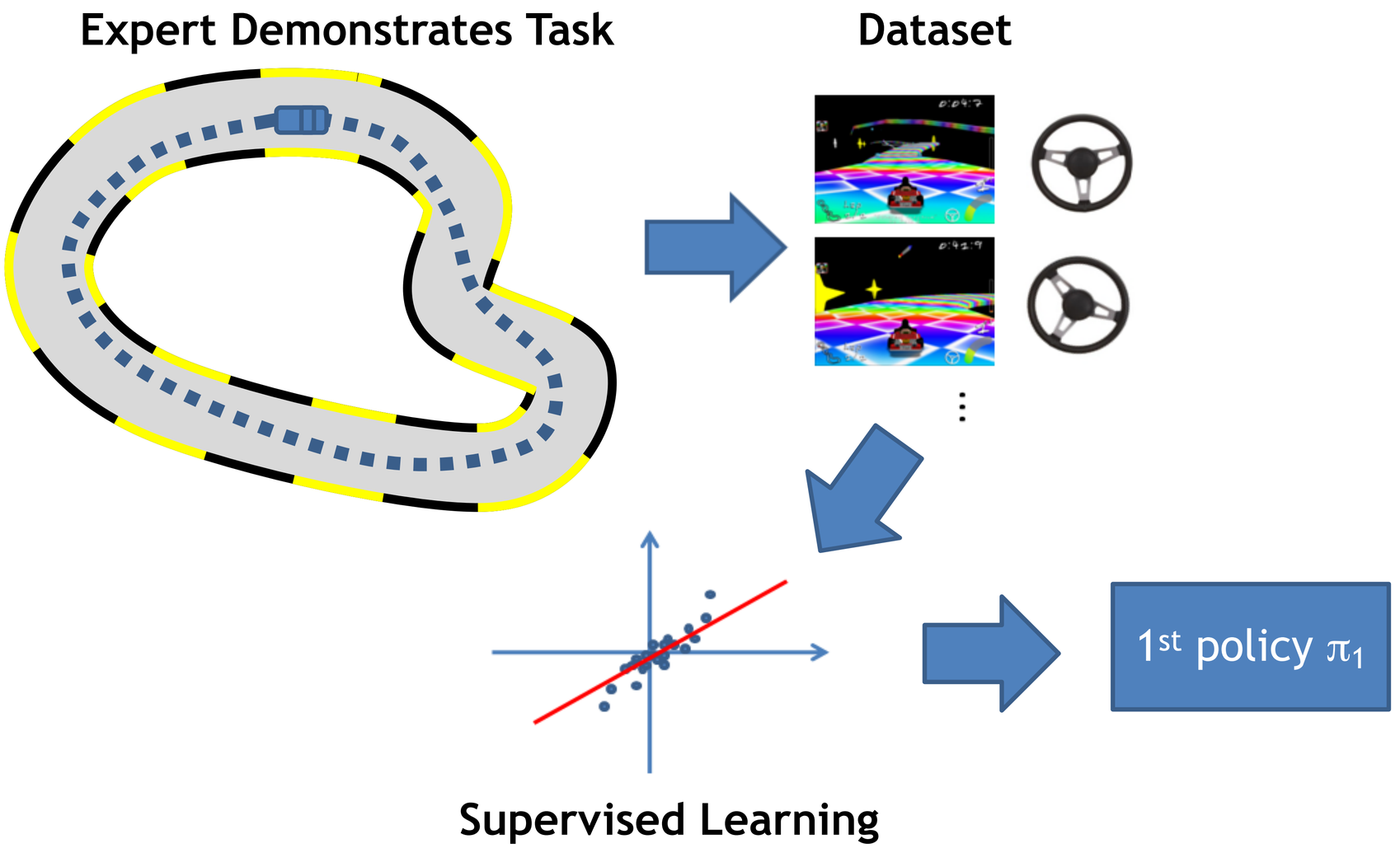


All DAgger slides credit: Drew Bagnell, Stephane Ross, Arun Venktraman

[Ross11a]

DAgger: Dataset Aggregation

0th iteration



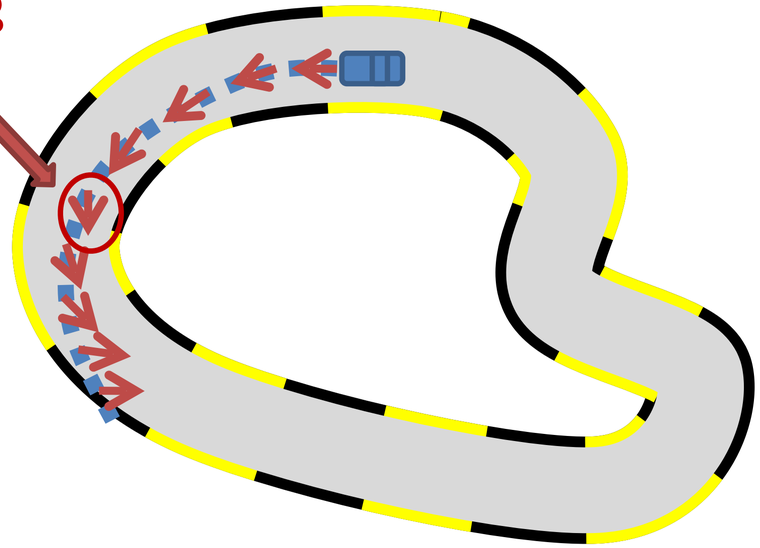
[Ross11a]

DAgger: Dataset Aggregation

1st iteration

Execute π_1 and Query Expert

Steering from expert



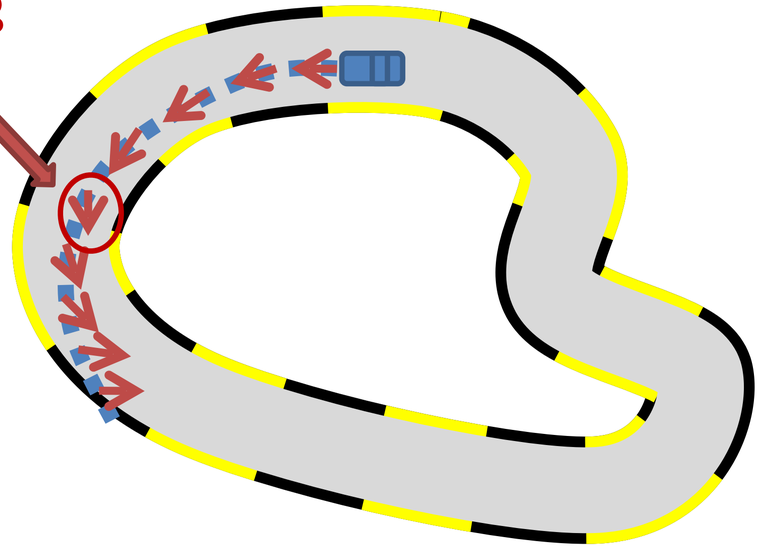
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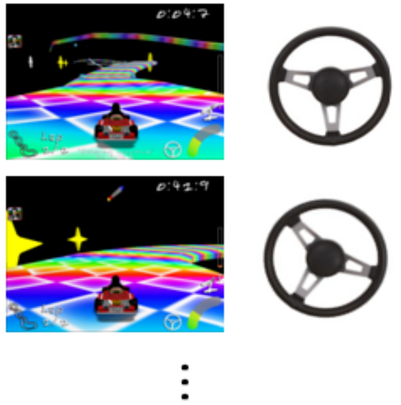
1st iteration

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New Data



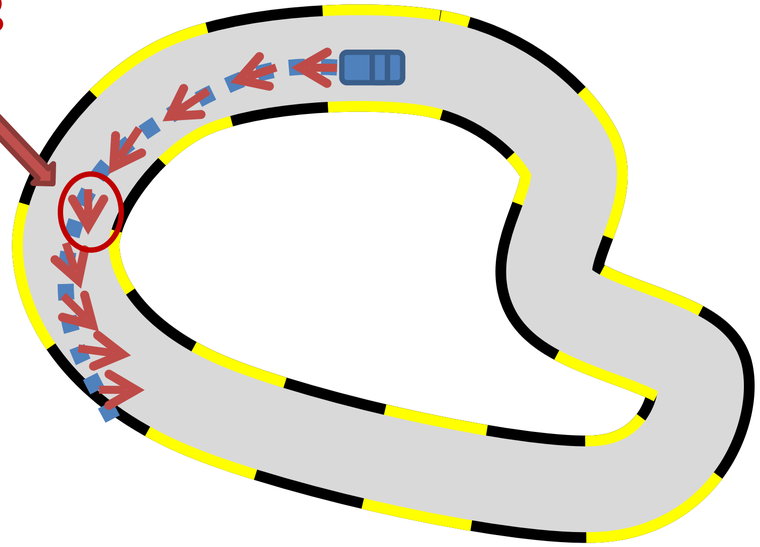
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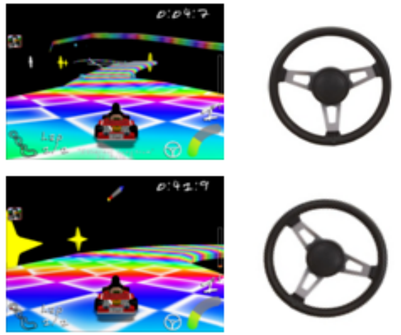
1st iteration

Execute π_1 and Query Expert

Steering from expert



New Data



States from the learned policy

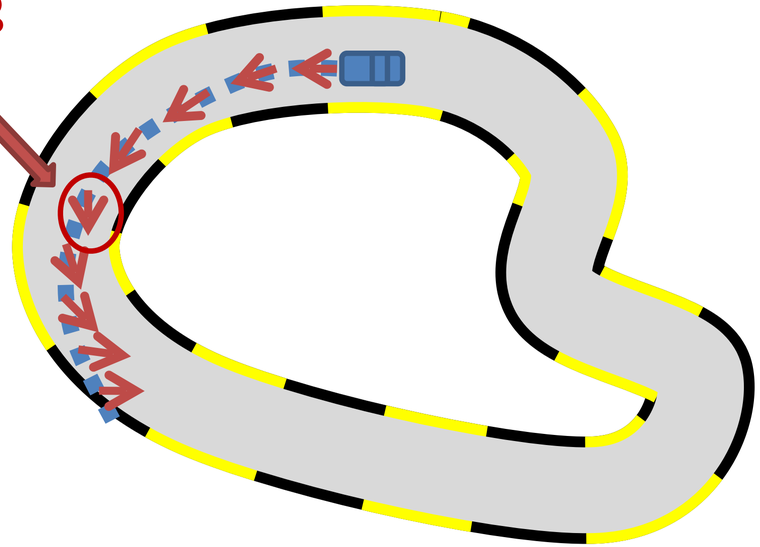
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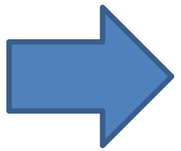
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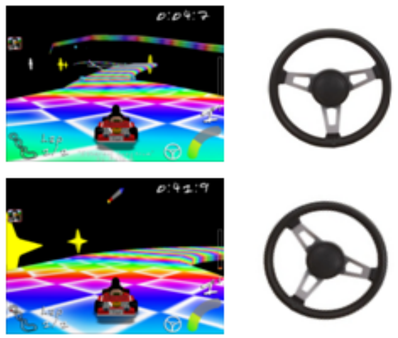
Execute π_1 and Query Expert

Steering from expert

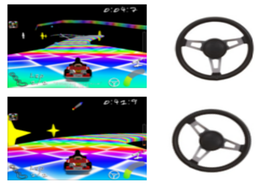




New Data



All previous data

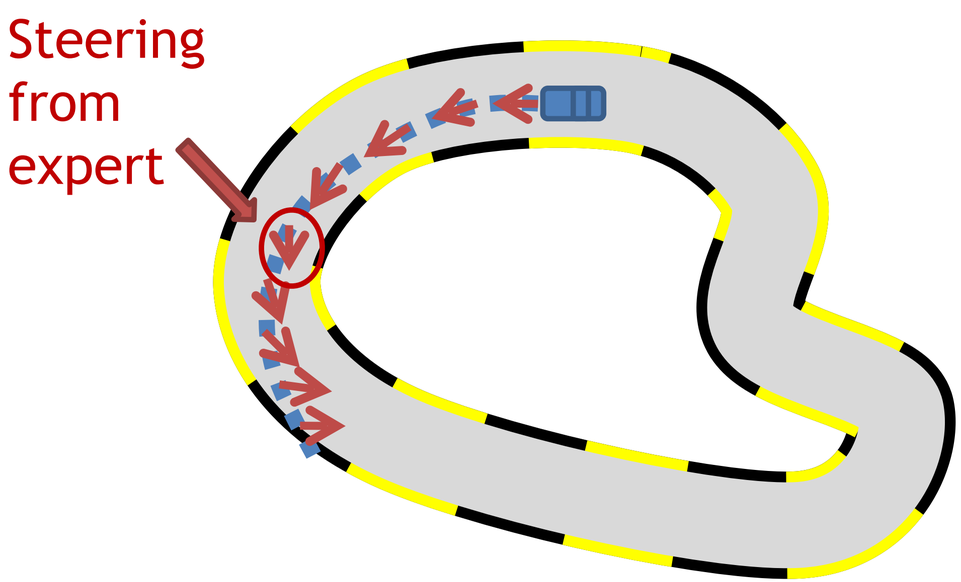


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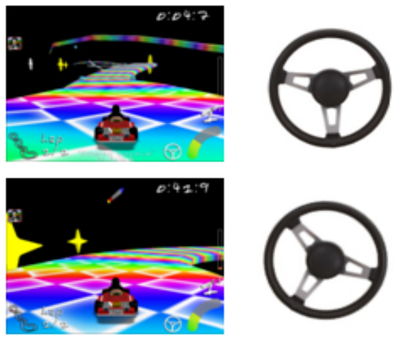
DAgger: Dataset Aggregation

1st iteration

Execute π_1 and Query Expert

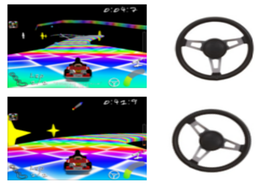


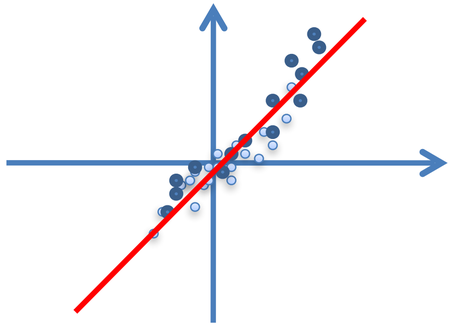
New Data



Aggregate
Dataset

All previous data





Supervised Learning

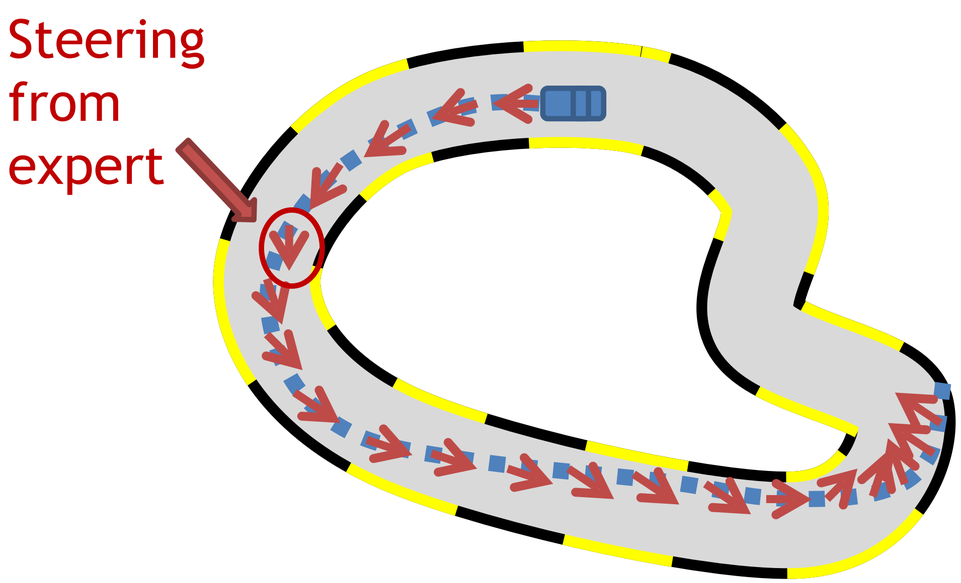
New policy
π_2

[Ross11a]

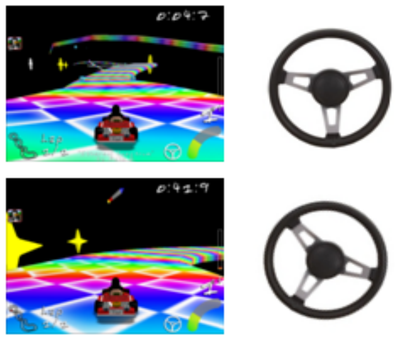
DAgger: Dataset Aggregation

2nd iteration

Execute π_2 and Query Expert

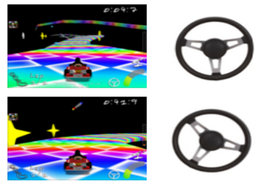


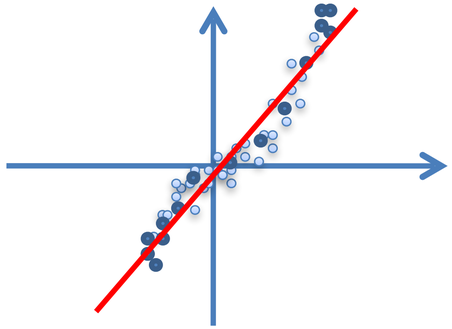
New Data



Aggregate
Dataset

All previous data





Supervised Learning

New policy
π_3

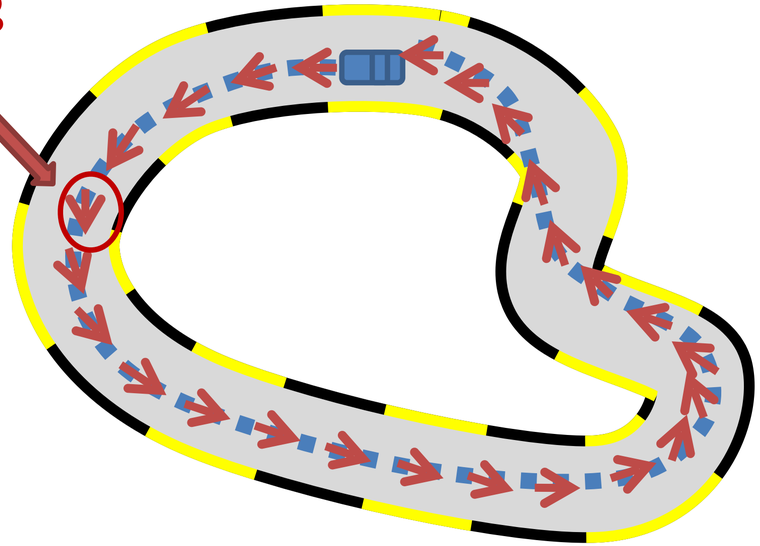
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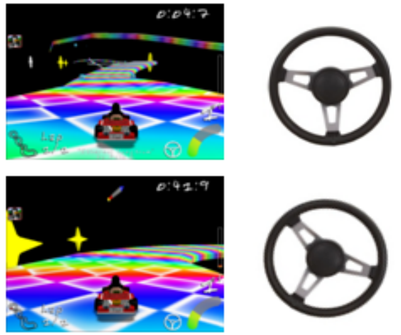
nth iteration

Execute π_{n-1} and Query Expert

Steering from expert

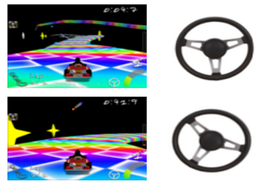


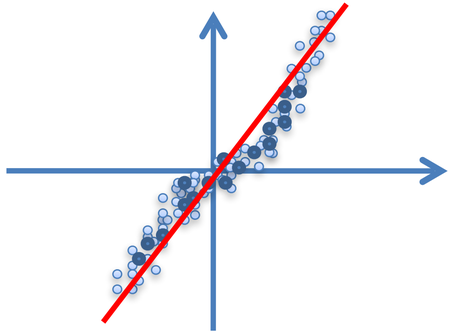
New Data



Aggregate Dataset

All previous data





Supervised Learning

New policy π_n

Success!

[Ross AISTATS 2011]



Success!

[Ross AISTATS 2011]



Success!

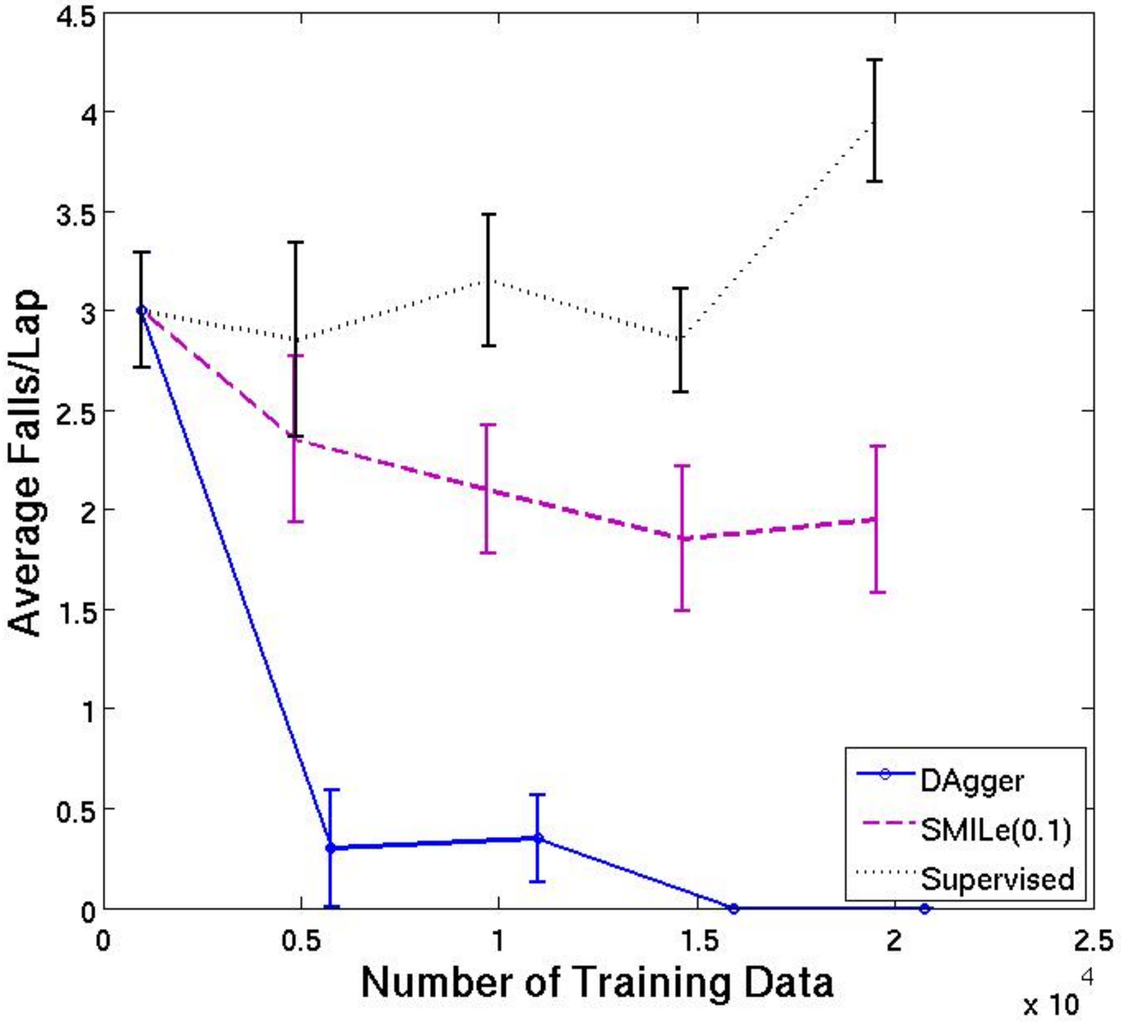
[Ross AISTATS 2011]



[Ross AISTATS 2011]

Average Falls/Lap

Better




FPS: 24
Attempt: 1 of 1
AgentLinear
Selected Actions:
RIGHT
SPEED


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Attempt: 1 of 1
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FPS: 24
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More fun than Video Games…



[Ross ICRA 2013]

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Forms of the Interactive Experts

Interactive Expert is expensive, especially when the expert is human…

But expert does not have to be human…

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[Pan et al, RSS 18, Best System Paper]



Fig. 4: The AutoRally car and the test track.

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Fig. 4: The AutoRally car and the test track.

Goal: learn a racing control policy that maps from data on cheap on-board sensors (raw-pixel imagine) to low-level control (steer and throttle)

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Goal: learn a racing control policy that maps from data on cheap on-board sensors (raw-pixel imagine) to low-level control (steer and throttle)



→ Steering + throttle

(a) raw image

Forms of the Interactive Experts

Example: high-speed off-road driving

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Forms of the Interactive Experts

Example: high-speed off-road driving
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Their Setup:
At Training, we have expensive sensors for accurate state estimation
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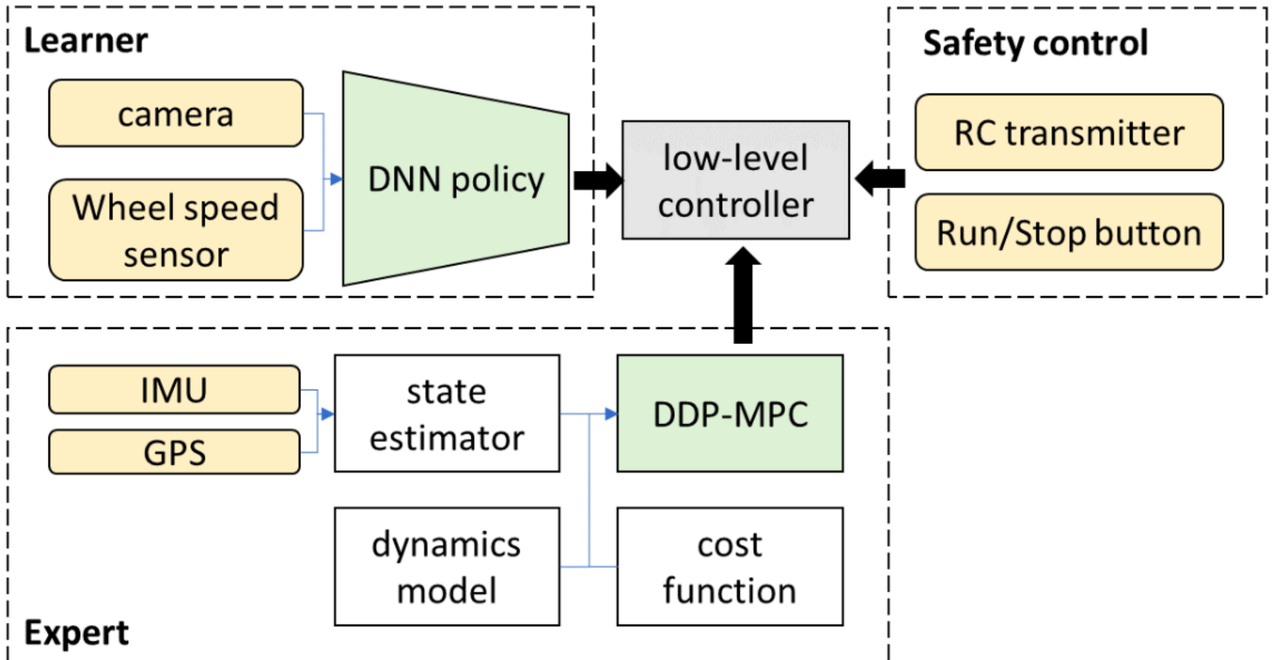
The MPC is the expert in this case!

Forms of the Interactive Experts

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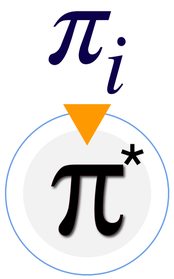
The MPC is the expert in this case!



Forms of the Interactive Experts

Named Entity Recognition

Input: After completing his Ph.D., Ellis worked at Bell Labs from 1969 to 1972 on probability theory..

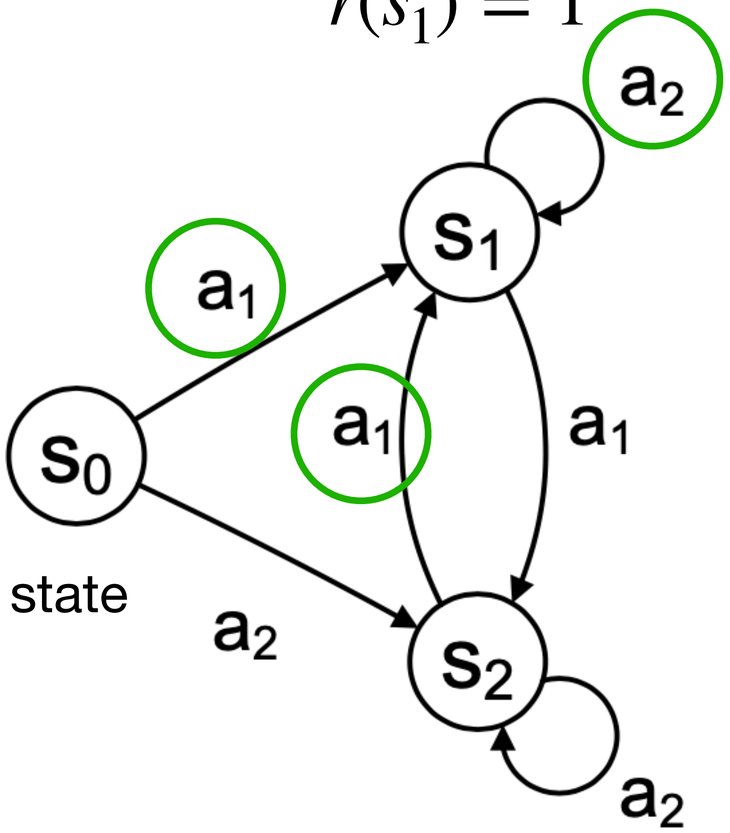


O O O O O PER O

O O O O O PER O O ORG ORG O O O O O O O

Distribution Shift: Example

$r(s_1) = 1$



$d_{s_0}^{\pi^\star}(s_0) = 1 - \gamma,\ d_{s_0}^{\pi^\star}(s_1) = \gamma,\ d_{s_0}^{\pi^\star}(s_2) = 0$

$$V_{s_0}^{\pi^\star} = \frac{\gamma}{1-\gamma}$$

Assume SL returned such policy $\widehat{\pi}$

$$\widehat{\pi}(s_0) = \begin{cases} a_1 & \text{w/ prob } 1 - \epsilon/(1-\gamma) \\ a_2 & \text{w/ prob } \epsilon/(1-\gamma) \end{cases}, \quad \widehat{\pi}(s_1) = a_2,\ \widehat{\pi}(s_2) = a_2$$

We will have good supervised learning error:

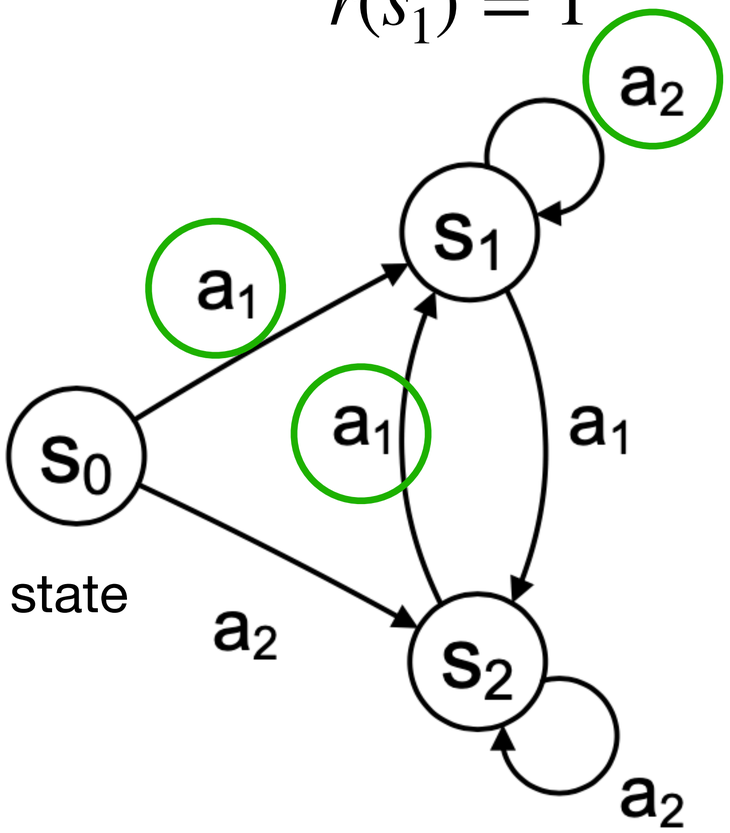
$$\mathbb{E}_{s \sim d_{s_0}^{\pi^\star}} \mathbb{E}_{a \sim \widehat{\pi}(\cdot|s)} \mathbf{1}\left(a \neq \pi^\star(s)\right) = \epsilon$$

But we have quadratic error in performance:

$$V_{s_0}^{\widehat{\pi}} = V_{s_0}^{\pi^\star} - \frac{\epsilon\gamma}{(1-\gamma)^2}$$

Distribution Shift: Example

$r(s_1) = 1$



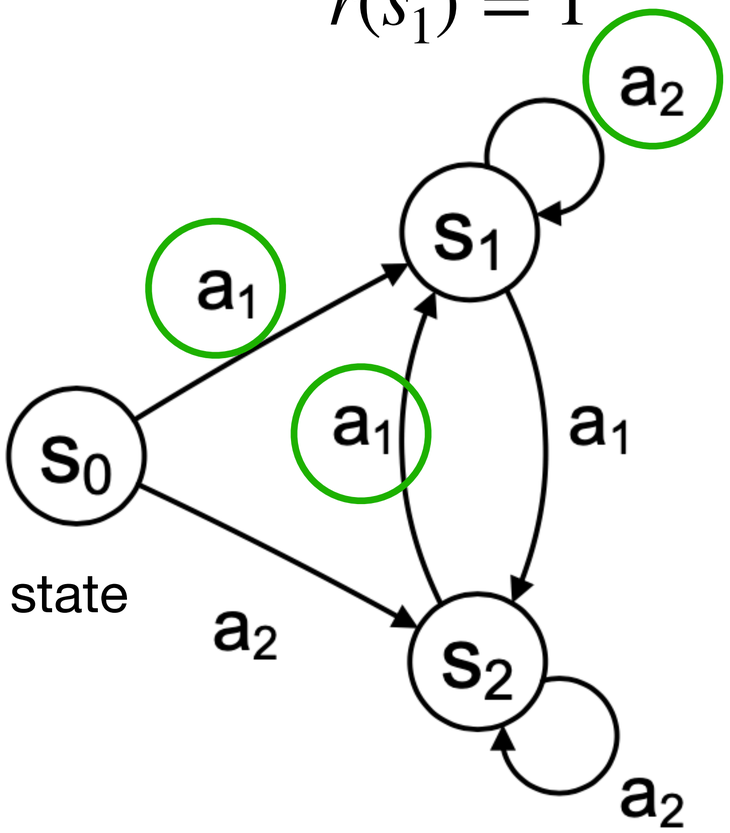
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Why DAgger can fix this problem?

$$d_{s_0}^{\pi^\star}(s_0) = 1 - \gamma, d_{s_0}^{\pi^\star}(s_1) = \gamma, d_{s_0}^{\pi^\star}(s_2) = 0$$

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Outline:

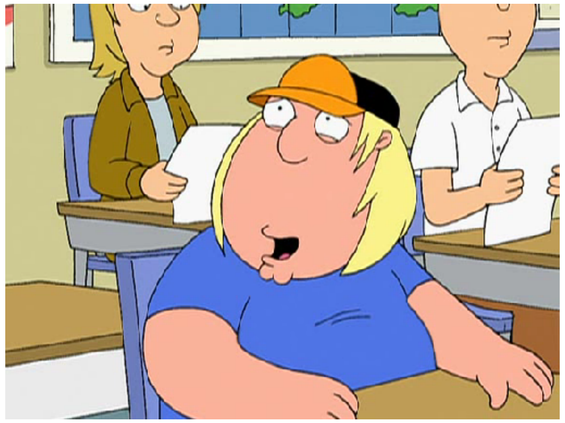


1. The DAgger (Data Aggregation) Algorithm

2. Analysis of DAgger: DAgger as online learning

Online Learning

Learner



convex Decision set Θ

Adversary

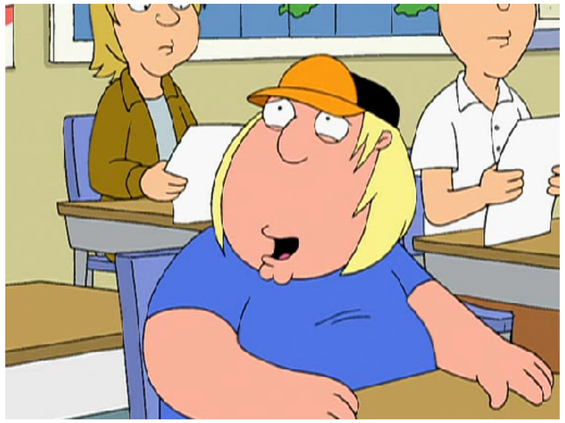


...

[Vovk92,Warmuth94,Freund97,Zinkevich03,Kalai05,Hazan06,Kakade08]

Online Learning

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convex Decision set Θ

Learner picks a decision θ_0



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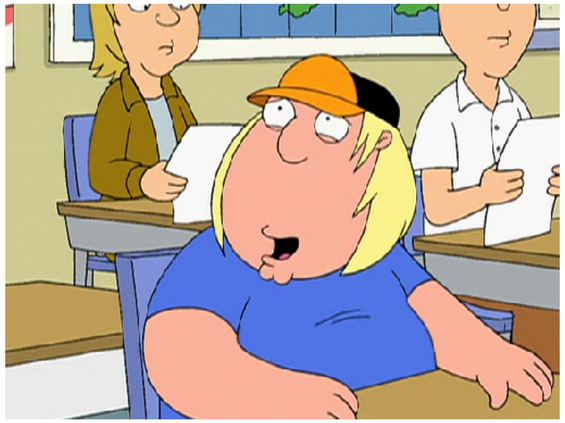


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Learner picks a decision θ_0 →

← Adversary picks a loss $\ell_0 : \Theta \to \mathbb{R}$

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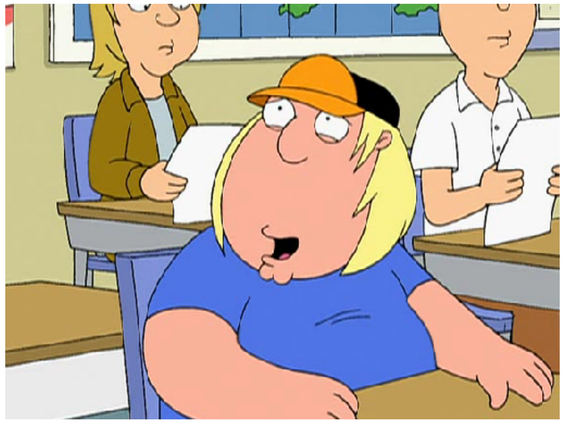


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Adversary picks a loss $\ell_0 : \Theta \to \mathbb{R}$ ←

Learner picks a new decision θ_1 →

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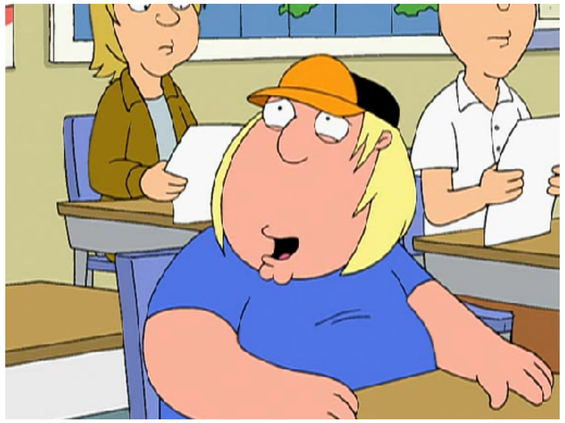
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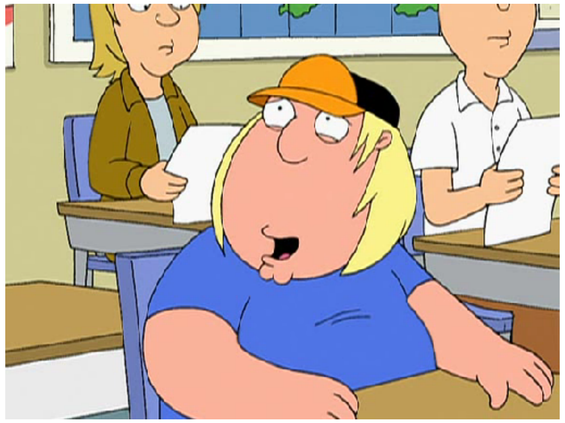
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$$\text{Regret} = \sum_{t=0}^{T-1} \ell_t(\theta_t) - \min_{\theta \in \Theta} \sum_{t=0}^{T-1} \ell_t(\theta)$$

[Vovk92,Warmuth94,Freund97,Zinkevich03,Kalai05,Hazan06,Kakade08]

Example: online linear regression

Can we perform linear regression online with non i.i.d (or even adversary) data?

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Adversary seems too powerful…

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$$\sum_{i=0}^{T-1} \ell_i(\theta_i) - \min_{\theta \in \text{Ball}} \sum_{i=0}^{T-1} \ell_i(\theta) = O\left(\sqrt{T}\right)$$

Generally, Follow-the-Regularized-Leader is no-regret

At time step t, learner has seen $\ell_0, \ldots \ell_{t-1}$, which new decision she could pick?

$$\textbf{FTRL: } \theta_t = \min_{\theta \in \Theta} \sum_{i=0}^{t-1} \ell_i(\theta) + \lambda R(\theta)$$

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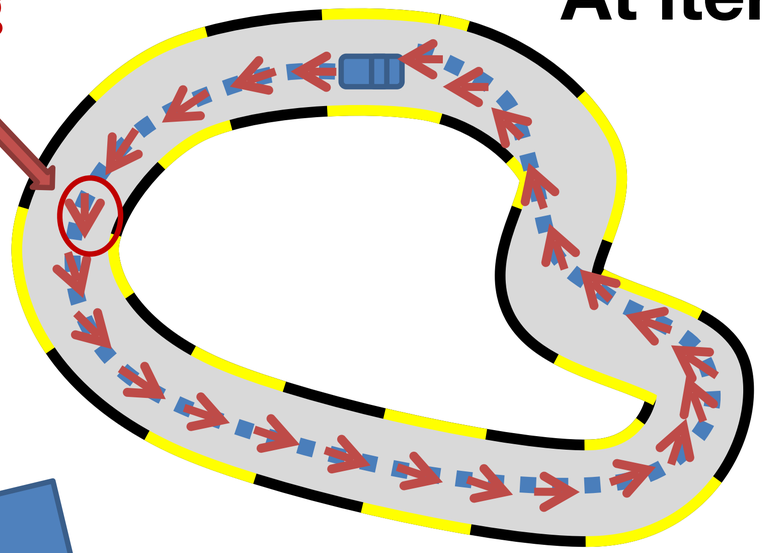
Informal Theorem (FTRL): when things are convex, FTRL is no-regret, i.e.,

$$\frac{1}{T}\left[\sum_{t=0}^{T-1} \ell_t(\theta_t) - \min_{\theta \in \Theta} \sum_{t=0}^{T-1} \ell_t(\theta)\right] = O\left(1/\sqrt{T}\right)$$

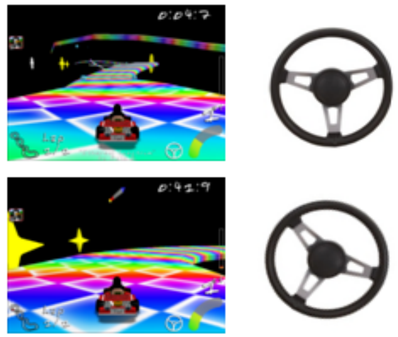
DAgger Revisit

At iteration t:

Steering from expert

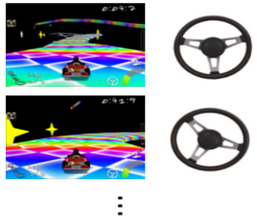


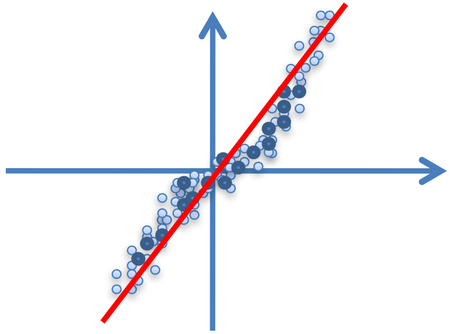
New Data



Aggregate Dataset

All previous data





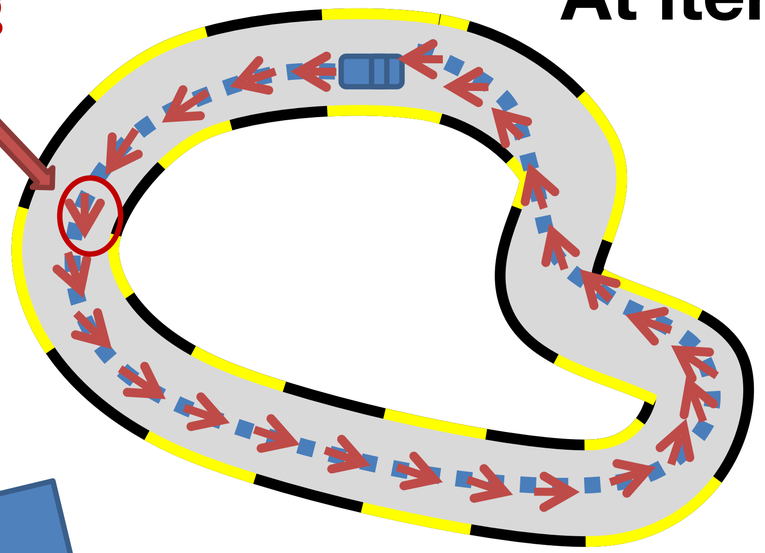
New policy π_n

Supervised Learning

DAgger Revisit

At iteration t:

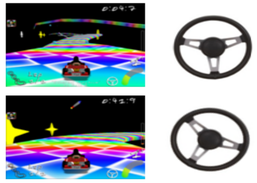
Steering from expert



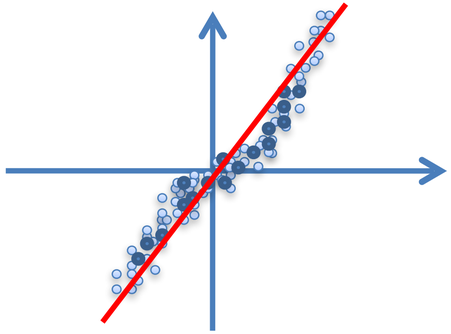
New Data

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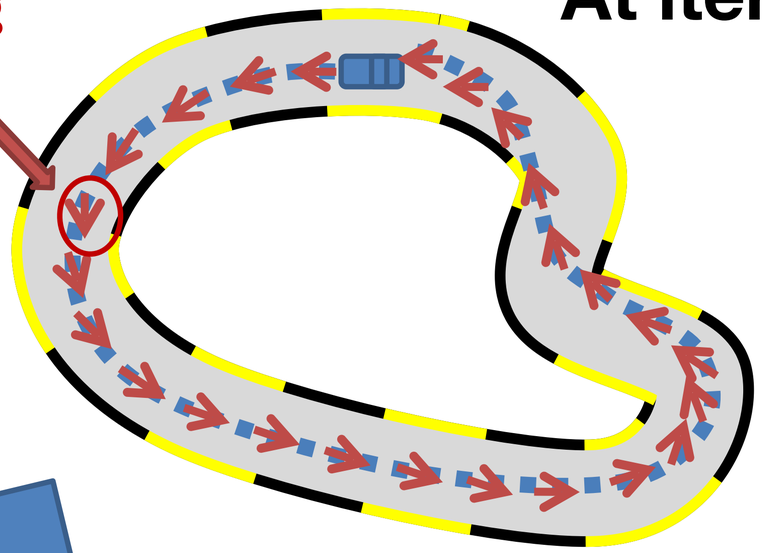


Supervised Learning

DAgger Revisit

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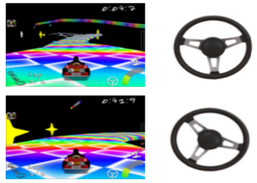


New Data

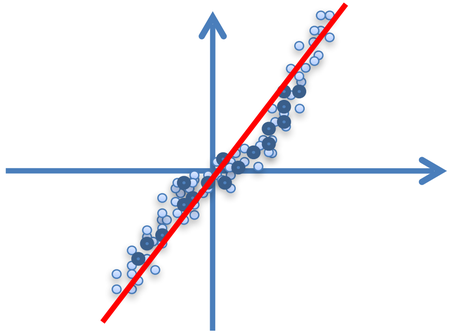
$$\ell_t(\pi) = \sum_{i=1}^{m} \|\pi(s^i) - \pi^\star(s^i)\|_2^2$$

Aggregate Dataset

All previous data



New policy π_n

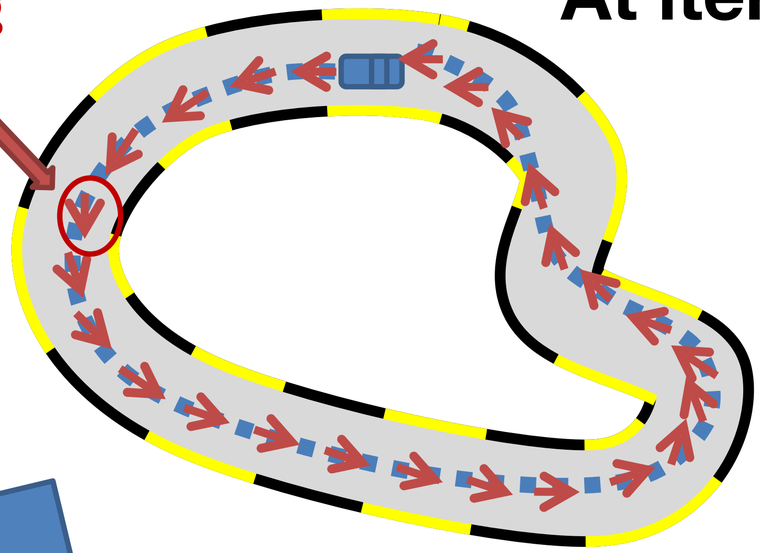


Supervised Learning

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At iteration t:

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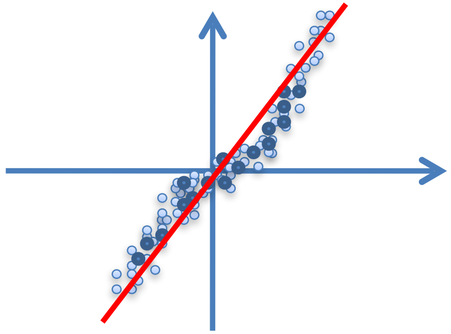


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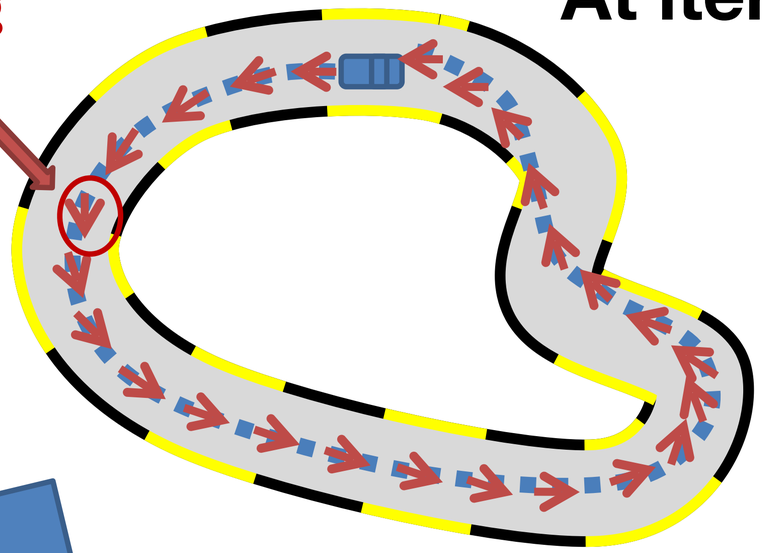
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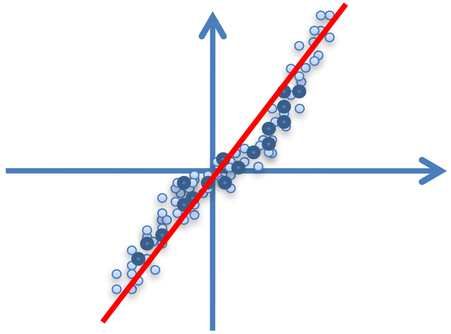
New Data

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Aggregate Dataset

All previous data

$$\sum_{i=0}^{t} \ell_i(\pi)$$



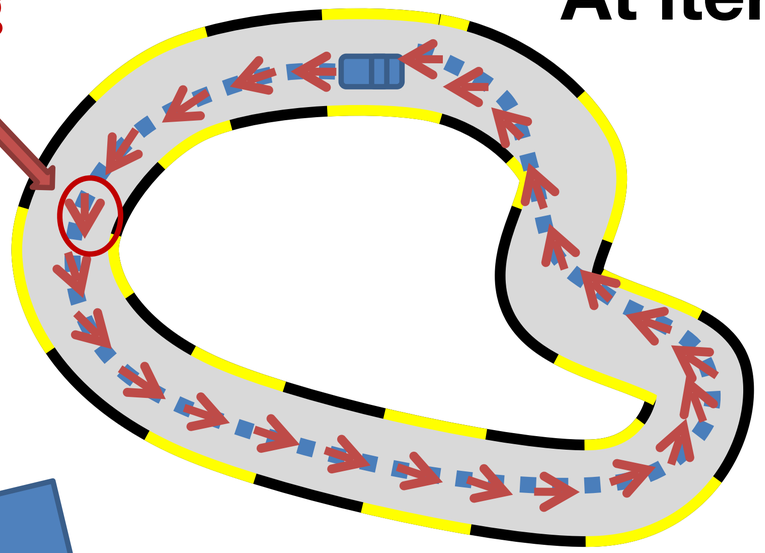
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Aggregate Dataset

All previous data

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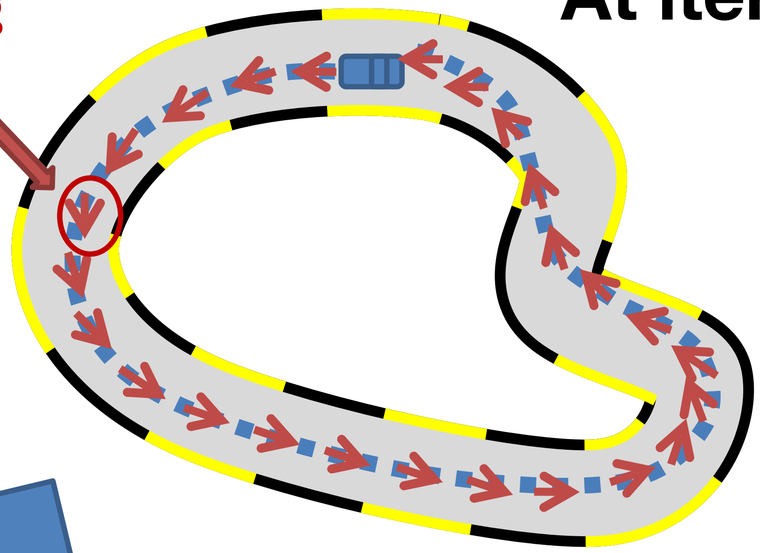
New policy π_n

Supervised Learning

DAgger Revisit

At iteration t:

Steering from expert



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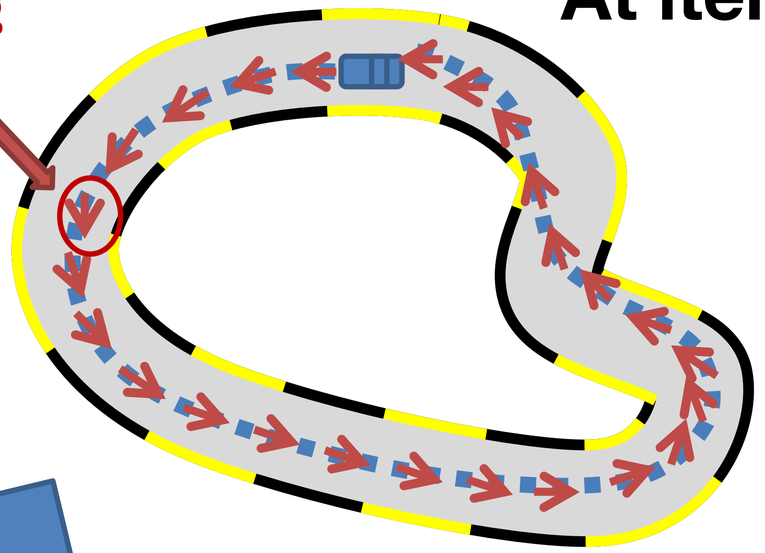
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Data Aggregation = Follow-the-Regularized-Leader Online Learner

Summary for Today

1. The DAgger algorithm

Initialize π^0, and dataset $\mathscr{D} = \varnothing$

For $t = 0 \rightarrow T - 1$:

1. W/ π^t, generate dataset $\mathscr{D}^t = \{s_i, a_i^\star\}, s_i \sim d_\mu^{\pi^t}, a_i^\star = \pi^\star(s_i)$
2. **Data aggregation**: $\mathscr{D} = \mathscr{D} + \mathscr{D}^t$
3. **Update policy via Supervised-Learning**: $\pi^{t+1} = \mathrm{SL}\left(\mathscr{D}\right)$

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2. We can see that DAgger is essentially an online-learning algorithm (FTRL)

Feedback and Attendance

https://bit.ly/4gYn2nC

